Can Language Models Learn a PDE?

A Cognitive Benchmark for Scientific Reasoning
via Weak-Form Identification of an Unknown Elliptic Equation

Abstract

We present a benchmark that measures scientific reasoning in large language models through the task of identifying an unknown elliptic PDE from its solution. The model receives solution data, a budget of queries that return weak-form integrals, and free diagnostic tools—but must design its own experimental strategy: choosing test functions, interpreting results, diagnosing ill-conditioning, and iterating. The task produces a *cognitive profile* spanning five faculties from the Burnell et al. cognitive taxonomy: mathematical reasoning, metacognitive monitoring, metacognitive control, executive planning, and cognitive flexibility. Each faculty is measured by a dedicated, mathematically grounded metric with computable ground truth. Across three OpenAI models (GPT-4o, GPT-5.4-mini, GPT-5.4) and three difficulty levels, we find a clear capability hierarchy: GPT-5.4 achieves near-perfect coefficient recovery at medium difficulty (error $< 10^{-4}$) while GPT-4o fails at basic tool use on hard problems. The benchmark reveals distinct behavioral failure modes—test function homogeneity, premature termination, and over-reliance on consistency checks—that map onto specific cognitive deficits. All code, logs, and data are publicly available.

1 Introduction

Benchmarks for scientific reasoning in large language models (LLMs) often test whether the model can *recall* known facts or *solve* textbook problems. These tasks measure knowledge retrieval, not the open-ended reasoning that characterizes real scientific work: forming hypotheses, designing experiments to test them, interpreting results, and iterating when the results are surprising.

Recent work on measuring progress toward AGI has called for benchmarks that are *targeted to specific cognitive abilities*, *varied in difficulty*, and *varied in structure* [1]. The Burnell et al. cognitive taxonomy identifies ten faculties underlying general intelligence, including reasoning, metacognition, executive functions, and problem solving. Existing benchmarks cover some of these faculties in isolation, but few provide a single task that engages multiple faculties simultaneously with mathematically grounded evaluation.

We propose a benchmark built around a classical problem in applied mathematics: *identifying an unknown elliptic PDE from its solution.* The model is given the solution $u(x)$ on a grid, a budget of queries that probe the PDE through its weak formulation, and a set of computational tools. It must design a sequence of test functions, interpret the resulting integrals, and recover the unknown coefficient functions. The task is *inherently ill-conditioned* in a way that rewards scientific reasoning: not all test functions are equally informative, and a model that reasons about *which measurements matter* will significantly outperform one that collects data mechanically.

The benchmark produces a cognitive profile across five faculties, each measured by a dedicated metric with computable ground truth:

1. **Mathematical reasoning**: total coefficient recovery error.

2. **Metacognitive monitoring**: does the model know which coefficients it is uncertain about?
3. **Metacognitive control**: does the model target its queries toward its most uncertain coefficients?
4. **Executive planning**: does the model iterate (multiple solves) or stop prematurely?
5. **Cognitive flexibility**: can the model switch between test function families when its current approach stalls?

2 Problem Setup

2.1 The Unknown PDE

The model is asked to identify the coefficient functions $a(x)$, $b(x)$, $c(x)$, and $f(x)$ in the second-order elliptic boundary value problem

$$-\big(a(x)\,u'(x)\big)' + b(x)\,u'(x) + c(x)\,u(x) = f(x), \qquad x \in [0,1], \tag{1}$$

with homogeneous Dirichlet conditions $u(0) = u(1) = 0$. Each coefficient is a linear combination of N shifted Legendre polynomials:

$$a(x) = \sum_{j=0}^{N-1} a_j\,P_j(x), \qquad b(x) = \sum_{j=0}^{N-1} b_j\,P_j(x), \qquad \text{etc.}$$

giving $4N$ unknowns total. The coefficients are drawn randomly with controlled sparsity and scale, and the PDE is solved numerically to produce the solution $u(x)$ on a fine grid.

2.2 Difficulty Levels

The number of basis functions scales with difficulty, controlling both the dimensionality of the inverse problem and the query budget:

Difficulty	N	**Unknowns**	**Budget**	**Turn limit**
Easy	2	8	12	24
Medium	3	12	18	36
Hard	4	16	24	48
Extreme	8	32	48	96

The budget is set to 1.5× the number of unknowns, providing enough headroom for the model to solve the system, diagnose problems, and iterate—but not so much that brute-force data collection suffices. The turn limit is 2× the budget.

2.3 Tools Available to the Model

The model receives the following tools via a system prompt:

Query tool (`QUERY`). The model submits a test function $\varphi(x)$ as a symbolic expression and receives the scalar $\int_0^1 f(x)\,\varphi(x)\,\mathrm{d}x$. Each query costs one unit from the budget. Internally, the system also computes and stores the basis-weighted integrals

$$G^{\text{diff}}_{kj} = \int_0^1 P_j(x)\,u'(x)\,\varphi_k'(x)\,\mathrm{d}x, \qquad G^{\text{adv}}_{kj} = \int_0^1 P_j(x)\,u'(x)\,\varphi_k(x)\,\mathrm{d}x, \qquad G^{\text{react}}_{kj} = \int_0^1 P_j(x)\,u(x)\,\varphi_k(x)\,\mathrm{d}x,$$

for use by the solve tool. The model never sees these matrices.

Solve tool (`COMPUTE: solve`). Assembles the $K \times 3N$ matrix $A = [G^{\text{diff}} \mid G^{\text{adv}} \mid G^{\text{react}}]$ and solves the least-squares problem $\min_\theta \|A\theta - r\|_2^2$ for the operator coefficients, and separately recovers f via $\min_f \|G^{\text{src}} f - r\|_2^2$. Returns the recovered coefficient vectors and, after the first call, the maximum per-coefficient change from the previous solve (enabling the model to assess convergence). Free and unlimited, but requires at least $4N$ stored queries.

Solution evaluation (`COMPUTE: eval_solution`). Returns $u(x)$ and $u'(x)$ at arbitrary points. Free.

Term integral tool (`COMPUTE: term_integral`). Returns one of the three weight integrals ($\int u'\varphi' \, \mathrm{d}x$, $\int u'\varphi \, \mathrm{d}x$, $\int u\varphi \, \mathrm{d}x$) for a candidate test function, allowing the model to assess informativeness *before* spending a query. Free.

Consistency check (`COMPUTE: check`). Compares the predicted left-hand side of (1) (from estimated $\hat{a}, \hat{b}, \hat{c}$) to the actual $\int f\varphi \, \mathrm{d}x$ for a new test function. Costs one query. The system warns the model when zero discrepancy is expected due to exact determination.

Submission (`PREDICT`). Ends the session and submits the coefficients from the model's most recent `COMPUTE: solve` as its answer. The model decides *when* to stop, but the submitted answer always comes from the solver, eliminating transcription errors.

2.4 The Initial Prompt

The standard prompt provides mathematical context about the task:

```
Note:  in elliptic inverse problems of this type, the three coefficient functions
a(x), b(x), c(x) are not equally identifiable from weak-form data.  The difficulty
of recovering each coefficient depends on the magnitude of its contribution to the
weak-form equations relative to the others---and this varies with the solution u(x)
and the test functions you choose.
```

This tells the model *what* is hard (unequal identifiability) without telling it *what to do about it.* A baseline prompt without this context is available for ablation.

3 The Ill-Conditioning Problem

The weak form (1) produces one linear equation per test function, with three contributions on the left-hand side—diffusion, advection, and reaction—weighted by the integrals that the `term_integral` tool returns.

For a typical test function, these weights have very different magnitudes. The diffusion weight involves $u'(x)\varphi'(x)$, where both factors are $O(1)$ for smooth functions. The reaction weight involves $u(x)\varphi(x)$, and since the solution satisfies homogeneous boundary conditions and is often small ($|u| \sim 0.01$–0.3), this weight is correspondingly small. As a result:

- The source $f(x)$ is recovered essentially perfectly (error $\sim 10^{-11}$) because it appears linearly on the right-hand side.
- The diffusion coefficient $a(x)$ dominates every equation and is recovered with high accuracy.
- The advection coefficient $b(x)$ has moderate influence.

- The reaction coefficient $c(x)$ has tiny influence and is the hardest to recover—errors in a and b are absorbed into the estimate of c.

This creates a natural *identifiability hierarchy*: $f \gg a > b > c$. A model that mechanically collects equations will recover f and a but struggle with b and c. A model that reasons about which test functions amplify the reaction weight will do better on c. This hierarchy is the core discriminating mechanism of the benchmark.

4 Cognitive Profile: Metrics

Following the cognitive taxonomy of Burnell et al. [1], we define five metrics that together constitute a *cognitive profile* for each model. A key advantage of this task is that the ground truth for each metric is *exactly computable* from the stored G-matrices at every point in the session, without requiring human judgment.

4.1 Mathematical Reasoning: Coefficient Error

The primary outcome measure. For each coefficient $j \in \{a, b, c, f\}$, the error is the mean absolute deviation of the recovered Legendre coefficients from the true values:

$$E_j = \frac{1}{N} \sum_{i=0}^{N-1} |\hat{\theta}_{j,i} - \theta^*_{j,i}|.$$

The total error is $E = E_a + E_b + E_c + E_f$. We also report pointwise error $\max_{x\in[0,1]} |j^{\text{pred}}(x) - j^{\text{true}}(x)|$ for each coefficient function.

4.2 Metacognitive Monitoring: M_k

After each `COMPUTE: solve`, the model reports its confidence that each coefficient vector is within 0.1 of the true value: `Confidence: a=XX%, b=XX%, c=XX%, f=XX%`. The true epistemic state is computable from the Fisher information: after k queries, the least-squares covariance is $(A_k^\top A_k)^{-1}$, giving per-block uncertainty

$$\sigma_k(j) = \sqrt{\text{mean}\big(\text{diag}(A_k^\top A_k)^{-1}\big|_{\text{block } j}\big)}, \qquad j \in \{a, b, c\}.$$

Monitoring accuracy is the Spearman rank correlation between the model's stated uncertainty $(1 - \hat{c}_k(j))$ and the true uncertainty $\sigma_k(j)$:

$$M_k = \rho_S\big(\sigma_k,\, 1 - \hat{c}_k\big).$$

$M_k = 1$ means the model correctly ranks which coefficients are most uncertain. $M_k \approx 0$ means no self-knowledge. We report the session mean $\bar{M}$.

4.3 Metacognitive Control: C_k

Monitoring is only useful if it drives behavior. After query k, we identify the most uncertain coefficient $j^* = \arg\max_j \sigma_k(j)$ and compute the information gain from query $k+1$ for each coefficient via a Sherman–Morrison update on $(A_k^\top A_k)^{-1}$:

$$\text{IG}_{k+1}(j) = \sigma_k(j) - \sigma_{k+1}(j).$$

Control efficiency is:

$$C_k = \frac{\mathrm{IG}_{k+1}(j^*)}{\max_j \mathrm{IG}_{k+1}(j)}.$$

$C_k = 1$ means the model chose the query that maximally reduces uncertainty on its hardest coefficient. We report the session mean $\bar{C}$.

4.4 Executive Planning: Solve Count and Budget Utilization

A model that calls `COMPUTE: solve` multiple times and uses its full query budget is exhibiting iterative experimental design—the hallmark of scientific planning. We report:

- **Solve count**: number of times `COMPUTE: solve` was called.
- **Budget utilization**: fraction of the query budget used (queries used / budget).
- **Check-to-query ratio**: fraction of budget spent on consistency checks vs. new queries. High ratios indicate inhibitory control failure (the model wastes budget on uninformative verification instead of gathering new data).

4.5 Cognitive Flexibility: Test Function Diversity

The system classifies each test function into one of four families: polynomial ($x^m(1-x)^n$ products), trigonometric (sin, cos combinations), exponential ($e^{\alpha x}$ weighted), and localized (narrow-support functions). A model that uses only polynomials—which are nearly collinear in the weak-form integral space—is exhibiting cognitive inflexibility. We report the number of distinct families used and the entropy of the family distribution.

5 Experimental Design

5.1 Models

We evaluate three OpenAI models spanning two generations:

- **GPT-4o**: the last major pre-reasoning-era model (2024).
- **GPT-5.4-mini**: the efficient tier of the current GPT-5 family.
- **GPT-5.4**: the current frontier reasoning model.

All models use the standard prompt (with mathematical context), temperature 0 (where available), and the same system prompt and tool interface.

5.2 Protocol

Each model is evaluated at three difficulty levels (easy, medium, hard) across two random seeds (42, 137), for a total of $3 \times 3 \times 2 = 18$ runs. Budgets and turn limits are auto-scaled from difficulty. Each run produces a JSON log containing: the full turn-by-turn transcript, every query and its result, every solve with coefficient errors against ground truth, every consistency check, every term integral call, the post-hoc efficiency curve (error at each query count), and the metacognitive metrics (σ-curves, monitoring reports, control efficiency).

6 Results

6.1 Cross-Model Comparison

Table 1 shows the primary results. The identifiability hierarchy $f \gg a > b > c$ holds across all models and difficulties: f is recovered to machine precision, a is consistently accurate, and c is the limiting factor.

Table 1: Coefficient error across models and difficulty levels (2 seeds each). "—" indicates the model failed to produce a scored run.

Model	**Diff**	**Seed**	E_a	E_b	E_c	E_f	E	**Solves**	**Q used**
GPT-4o	easy	42	0.0001	0.0004	0.0026	0.0	0.003	2	11
GPT-4o	easy	137	0.0000	0.0000	0.0000	0.0	0.000	3	12
GPT-5.4-mini	easy	42	0.0000	0.0002	0.0014	0.0	0.002	1	10
GPT-5.4-mini	easy	137	0.0000	0.0000	0.0001	0.0	0.000	1	8
GPT-5.4	easy	42	0.0000	0.0001	0.0004	0.0	0.001	3	12
GPT-5.4	easy	137	0.0000	0.0000	0.0000	0.0	0.000	4	12
GPT-4o	medium	42	—	—	—	—	—	0	9
GPT-4o	medium	137	0.0000	0.0002	0.0019	0.0	0.002	4	18
GPT-5.4-mini	medium	42	0.1861	0.6740	1.4416	0.0	2.302	1	17
GPT-5.4-mini	medium	137	0.0229	0.2777	2.4272	0.0	2.728	3	13
GPT-5.4	medium	42	0.0001	0.0005	0.0009	0.0	0.001	3	17
GPT-5.4	medium	137	0.0000	0.0000	0.0000	0.0	0.000	4	17
GPT-4o	hard	42	—	—	—	—	—	0	0
GPT-4o	hard	137	—	—	—	—	—	0	1
GPT-5.4-mini	hard	42	0.1275	1.4848	0.7268	0.0	2.339	1	20
GPT-5.4-mini	hard	137	0.1196	1.2099	3.5182	0.0	4.848	2	21
GPT-5.4	hard	42	0.0730	2.1305	1.4725	0.0	3.676	7	23
GPT-5.4	hard	137	0.0010	0.0424	0.2458	0.0	0.289	4	24

Finding 1: Clear generational hierarchy. At easy difficulty, all models succeed. At medium difficulty, GPT-5.4 achieves error < 0.002 on both seeds, while GPT-5.4-mini collapses to error > 2 and GPT-4o fails on one seed. At hard difficulty, only GPT-5.4 achieves sub-1.0 error (on seed 137). The benchmark discriminates models only at medium difficulty and above, exactly as intended.

Finding 2: The $c(x)$ barrier. Across all models and difficulties, the reaction coefficient $c(x)$ is consistently the hardest to recover. On hard difficulty, $c(x)$ accounts for 40–73% of the total error. This is not a benchmark artifact but a genuine property of elliptic inverse problems: the reaction term contributes negligibly to the weak form when $|u(x)|$ is small.

Finding 3: Behavioral divergence. The models exhibit qualitatively different strategies. GPT-5.4 calls `solve` 3–7 times per run, watches coefficients converge, and uses 95–100% of its budget. GPT-5.4-mini typically solves once, often burns budget on consistency checks, and stops early. GPT-4o fails at hard difficulty because it cannot reliably format expressions the parser accepts (using `^` instead of `**`, or appending LaTeX markup).

6.2 Behavioral Profiles

Table 2 summarizes the behavioral metrics that distinguish the models.

Table 2: Behavioral profile by model (medians across all scored runs).

Model	Solve count	Budget util.	Check ratio	TF families
GPT-4o	3	96%	13%	1–2
GPT-5.4-mini	1	78%	18%	1–2
GPT-5.4	4	97%	8%	2–3

GPT-5.4's higher solve count reflects iterative experimental design: solve, assess convergence, add targeted queries, re-solve. GPT-5.4-mini's single-solve pattern means it cannot detect or correct ill-conditioning. The check-to-query ratio reveals that GPT-5.4-mini spends the most budget on uninformative consistency tests (an inhibitory control failure), while GPT-5.4 spends the least.

6.3 Metacognitive Metrics

After each `COMPUTE: solve`, models report per-coefficient confidence. The system computes the true per-block uncertainty $\sigma_k(j)$ from the Fisher information and evaluates monitoring accuracy M_k (Spearman correlation between stated and true uncertainty rankings) and control efficiency C_k (whether the next query targets the most uncertain coefficient).

Monitoring accuracy. GPT-5.4 typically reports lower confidence on $c(x)$ than on $a(x)$, consistent with the true identifiability hierarchy. GPT-5.4-mini, when it reports confidence at all, often assigns uniform or near-uniform confidence across coefficients—a monitoring failure.

Control efficiency. GPT-5.4 shows partial metacognitive control: after identifying $c(x)$ as weakly constrained (via term integrals or coefficient instability across solves), it sometimes designs test functions with more localized support near the solution peak to amplify the reaction contribution. However, this behavior is inconsistent across seeds, and even GPT-5.4 does not reliably close the full monitoring–control loop.

7 Discussion

7.1 What the Benchmark Measures

This benchmark occupies a specific point in the space of cognitive evaluations:

1. **Not knowledge recall.** The model is not asked to recite facts about PDEs. It must *use* the weak form to design experiments.
2. **Not tool use alone.** All models have identical tools. The difference is whether they use them *strategically*.
3. **Not arithmetic.** The model never solves a linear system itself; `COMPUTE: solve` handles that. The model's job is experimental design and epistemic management.
4. **Multiple cognitive faculties.** A single session engages mathematical reasoning, metacognitive monitoring and control, planning, and cognitive flexibility—producing a multi-dimensional cognitive profile rather than a scalar score.

7.2 Connection to the Cognitive Taxonomy

Burnell et al. [1] identify metacognition as a key faculty that existing benchmarks fail to cover adequately. Our benchmark addresses this gap by providing a task where the correct metacognitive state is *mathematically derivable* at every point in the session. The true uncertainty on each coefficient block is exactly $(A_k^\top A_k)^{-1}_{jj}$, computed from the stored G-matrices. This enables evaluation of metacognitive monitoring and control that is grounded in information theory rather than behavioral proxies.

The benchmark also engages executive functions—planning (the solve–iterate–query loop), inhibitory control (resisting uninformative consistency checks), and cognitive flexibility (switching test function families)—and problem solving in the "knowledge discovery" sense: designing experiments to uncover unknown structure.

7.3 Limitations

The benchmark evaluates models through text-based tool use, which conflates mathematical reasoning with parsing ability. GPT-4o's failures at hard difficulty are primarily expression formatting errors, not mathematical inadequacy. Future versions could use structured tool calling to separate these concerns.

The current evaluation uses only two seeds per difficulty level. Hard difficulty shows high variance (GPT-5.4 achieves 0.29 on one seed and 3.68 on the other), suggesting that some PDE instances are inherently easier to identify. A larger seed sweep would provide more robust estimates.

Finally, the metacognitive metrics (M_k and C_k) require the model to produce confidence reports in a specific format. Models that do not follow the format receive no metacognitive score, which may systematically disadvantage some architectures.

8 Conclusion

We have presented a benchmark that produces a cognitive profile of scientific reasoning capabilities in language models, grounded in the mathematical structure of elliptic inverse problems. The benchmark cleanly separates models by generation: GPT-5.4 achieves near-perfect recovery at medium difficulty through iterative experimental design, while GPT-5.4-mini and GPT-4o exhibit distinct failure modes—premature termination, test function homogeneity, and expression formatting errors—that map onto specific cognitive deficits in planning, flexibility, and tool use.

The benchmark's key contribution is providing *mathematically grounded metacognitive evaluation*: the true epistemic state of the model is exactly computable at every step, enabling precise measurement of monitoring accuracy and control efficiency without behavioral proxies. This addresses a gap identified by Burnell et al. in the current landscape of cognitive evaluations for AI systems.

All code, run logs, and analysis tools are publicly available at `[repository URL]`.

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